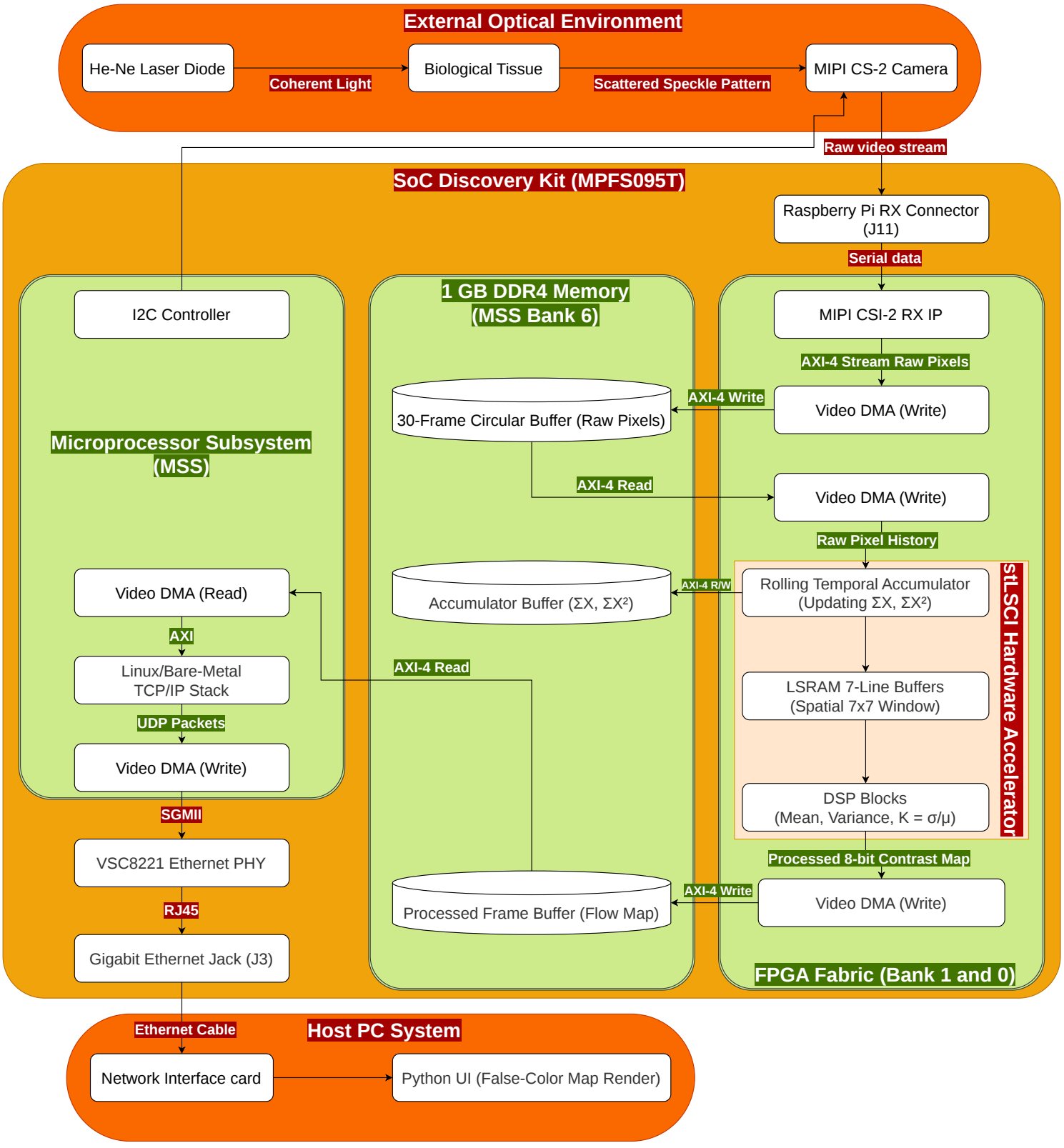

External Optical Environment
He-Ne Laser Diode
Coherent Light
Biological Tissue
Scattered Speckle Pattern
MIPI CS-2 Camera
Raw video stream
SoC Discovery Kit (MPFS095T)
Raspberry Pi RX Connector (J11)
Serial data
I2C Controller
1 GB DDR4 Memory (MSS Bank 6)
MIPI CSI-2 RX IP
AXI-4 Stream Raw Pixels
AXI-4 Write
30-Frame Circular Buffer (Raw Pixels)
Video DMA (Write)
Microprocessor Subsystem (MSS)
AXI-4 Read
Video DMA (Write)
Raw Pixel History
Video DMA (Read)
AXI-4 R/W
Accumulator Buffer (ΣX, ΣX²)
Rolling Temporal Accumulator (Updating ΣX, ΣX²)
stLSCI Hardware Accelerator
AXI
Linux/Bare-Metal TCP/IP Stack
AXI-4 Read
LSRAM 7-Line Buffers (Spatial 7x7 Window)
UDP Packets
Video DMA (Write)
DSP Blocks (Mean, Variance, K = σ/μ)
SGMII
Processed 8-bit Contrast Map
VSC8221 Ethernet PHY
AXI-4 Write
Processed Frame Buffer (Flow Map)
Video DMA (Write)
RJ45
Gigabit Ethernet Jack (J3)
FPGA Fabric (Bank 1 and 0)
Ethernet Cable
Host PC System
Network Interface card
Python UI (False-Color Map Render)